

UBBX (UB Book Exchange)

CSE 426 Term Project Documentation

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Live Project	https://ubbx.purusoni.com
Repository	https://github.com/purusoni/UBBX
UBBX Contract	0x780ef830cc76AD6E88FCe62A0e0485b68c227C41
Yoda Token	0xbd27d0b7F9fedb5A2A2C3ceF5dC9c70f3CF64Af2

Project Overview

UBBX is a decentralized textbook marketplace on Ethereum Sepolia. Listings are priced in Yoda (ERC-20) and participants transact directly through a wallet-connected web interface.

Motivation

University students frequently accumulate course textbooks that have little value to them after completing a class, while students entering those same courses face high retail and buyback prices. UBBX connects these two groups peer-to-peer, using a smart contract as the neutral intermediary for payment settlement and record-keeping.

Design Details

Smart Contract

The core marketplace logic is in `contracts/UBBX.sol`.

- `listBook(isbn, priceYoda, condition, metadataHash)` registers a listing and emits `BookListed`.
- `purchaseBook(id)` transfers Yoda from buyer to seller via `transferFrom`, marks the listing sold, and emits `BookPurchased`.

Data Design

On-chain state stores the minimum needed for trustless settlement: ISBN, price, condition, availability, and seller address. Descriptive content (title, author, edition, notes, image, contact) is stored off-chain in `metadata/<listingId>.json`. An optional `metadataHash` field allows buyers to verify catalog integrity on-chain.

Front End

The web interface is a static site served from the repository root (`index.html`, `app.js`), integrated with MetaMask and ethers.js v5. Listings are browsable without a wallet connection; listing and purchasing require one.

Development Details

Component	Files
Contract	contracts/UBBX.sol
Front end	index.html, app.js
ABIs	UBBX.json, yoda.json
Metadata	metadata/*.json

Verified Sepolia transactions:

Type	Truncated hash	Fee (ETH)
Deploy	0x3cd90f7473cbc6...	0.001623567
List Book	0x4e8d3b2e7bb379...	0.000243504
Purchase Book	0x8783668c41563f...	0.000108770
List Book	0xc74592d99d8189...	0.000222612
Purchase Book	0x17b01343ffe345...	0.000083120
List Book	0x9214c5506645c1...	0.000222612

How to Run the Code

Local Test

Serve the repository root over HTTP and open it in a browser:

```
python3 -m http.server 8080
```

Connect MetaMask and switch to Sepolia.

Deployment

1. Compile and deploy `contracts/UBBX.sol` (Solidity ^0.8.28) with the Yoda token address as the constructor argument.
2. Set `ubbxAddress` and `yodaAddress` in `app.js`.
3. Publish the static site via GitHub Pages (branch: `main`, folder: `root`).
4. For each listing with a metadata hash, commit the corresponding `metadata/<id>.json` file and push.



Figure 1: Remix IDE showing UBBX contract deployed at 0x780...27C41 on Sepolia.

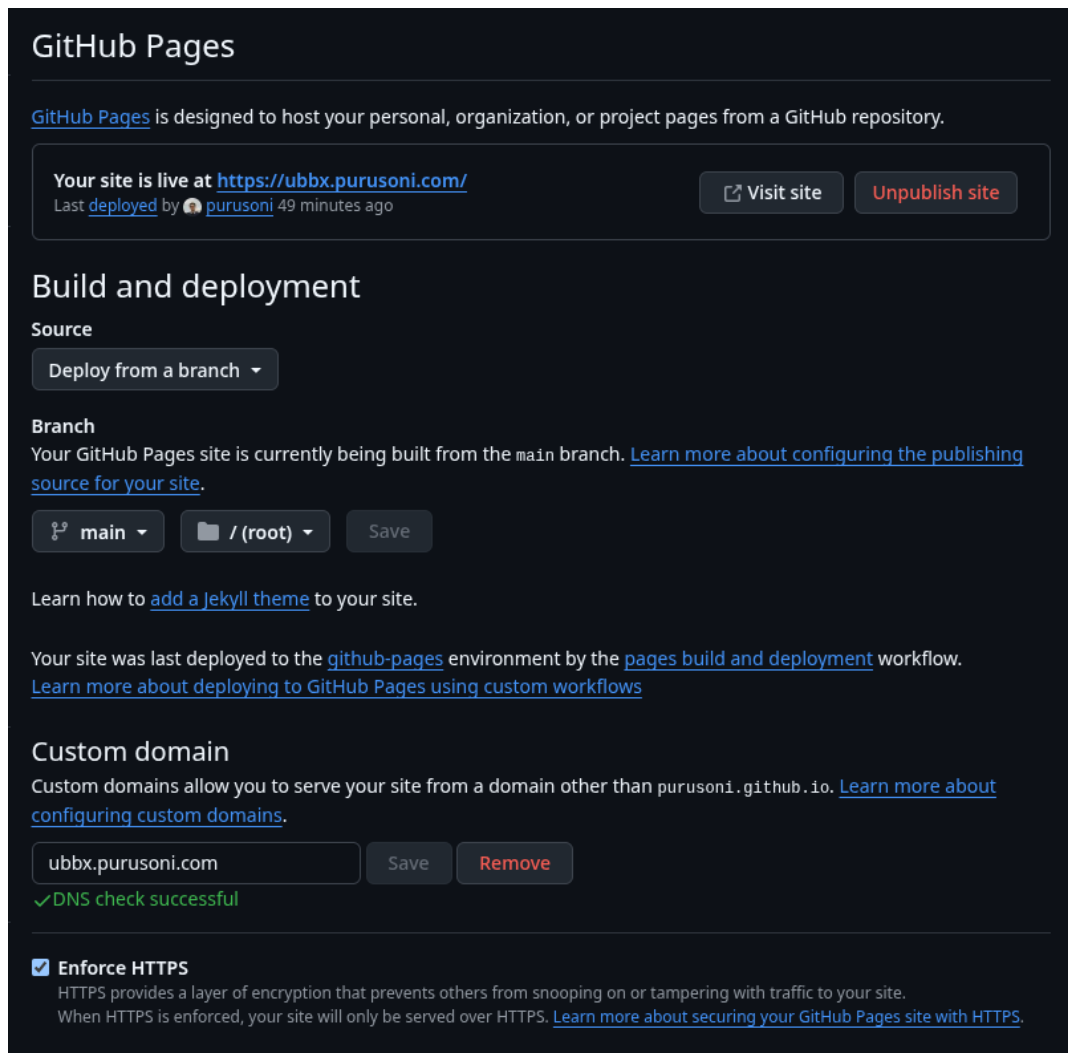


Figure 2: GitHub Pages configuration: `main` branch, root folder, custom domain `ubbx.purusoni.com`, HTTPS enforced.

User Instructions

1. Open <https://ubbx.purusoni.com> and connect MetaMask on Sepolia.
2. To list a book: enter ISBN, Yoda price, condition, and optional metadata fields, then submit.
3. To buy: select a listing from the catalog, enter its ID in the buy panel, approve Yoda spending if prompted, then confirm the purchase.
4. Transaction confirmation and listing status updates appear in the UI.

Wallets used in testing:

- Seller: `0xC6E031b6bd83891efFe6bAA15d7F7Df11549E9e1`
- Buyer: `0x59Fb57c7D932d1a61915df7C095Def91627567fa`

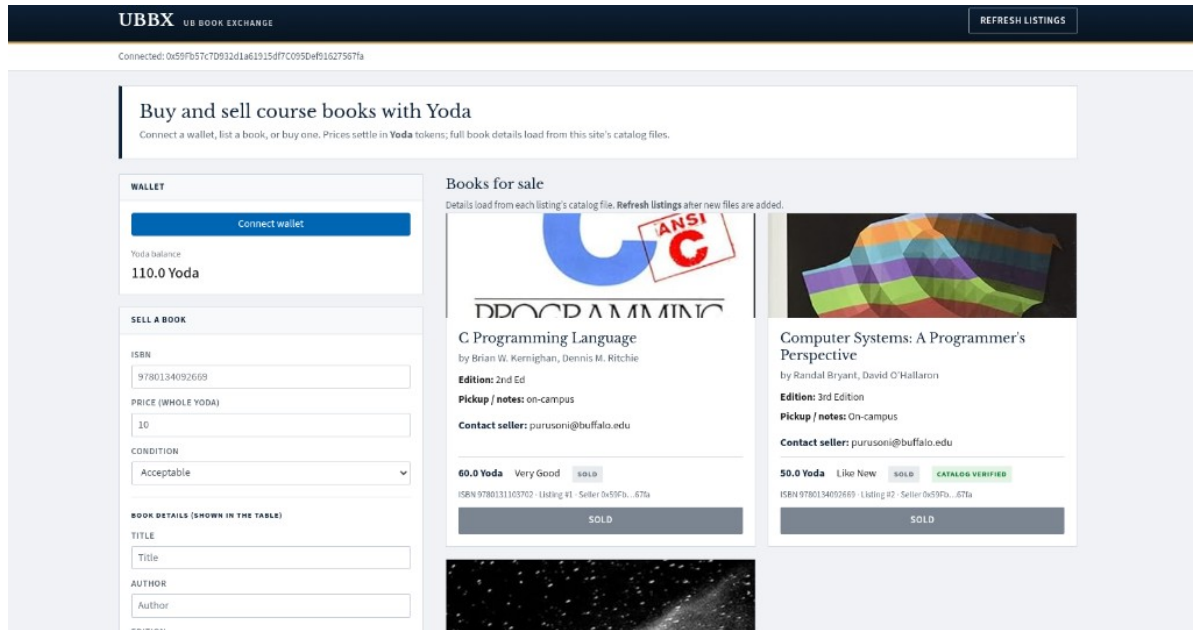


Figure 3: UBBX marketplace showing listed books, Yoda balance, and sell form.

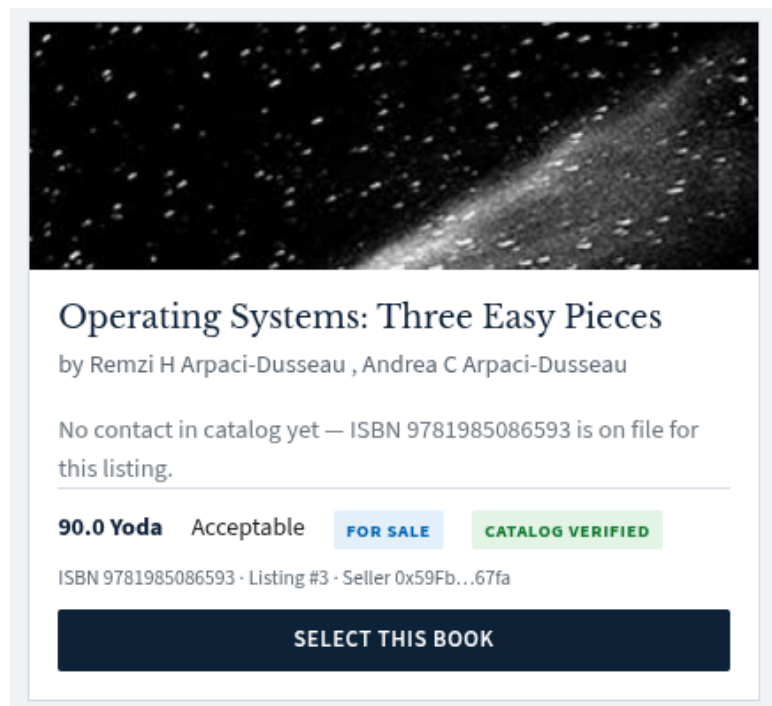


Figure 4: Individual listing card: price, condition, and catalog verification badge.

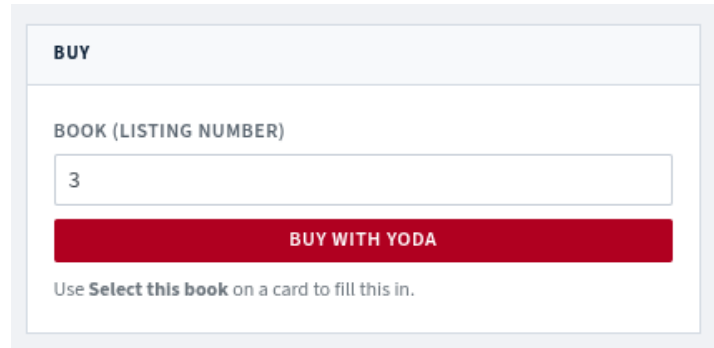


Figure 5: Buy panel with listing ID pre-filled from catalog selection.

Performance Metrics

Metric	Value
Total contract transactions	6 (1 deploy, 3 list, 2 purchase)
ERC-20 token transfer events	3
Avg. fee (non-deploy, 5 tx)	0.000176 ETH

Data sources:

- <https://sepolia.etherscan.io/address/0x780ef830cc76AD6E88FCe62A0e0485b68c227C41>
- <https://sepolia.etherscan.io/address/0xC6E031b6bd83891efFe6bAA15d7F7Df11549E9e1#token txns>
- <https://sepolia.etherscan.io/address/0x59Fb57c7D932d1a61915df7C095Def91627567fa#token txns>

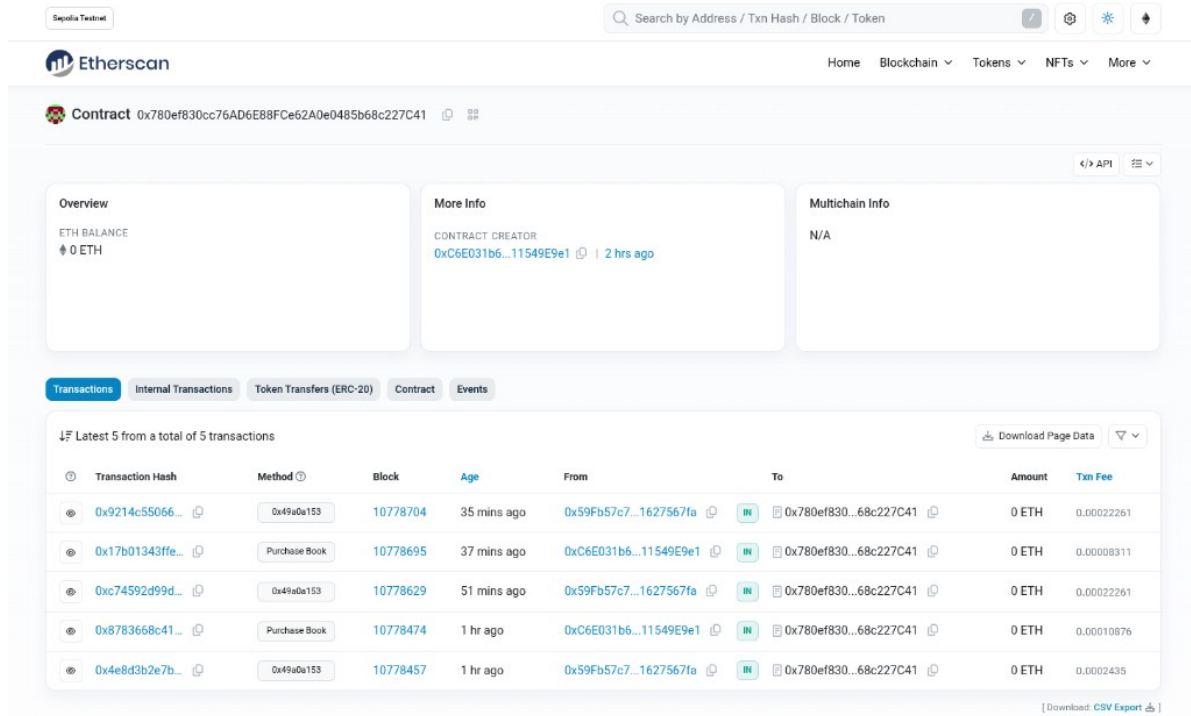


Figure 6: Contract transaction history on Sepolia with gas fees.

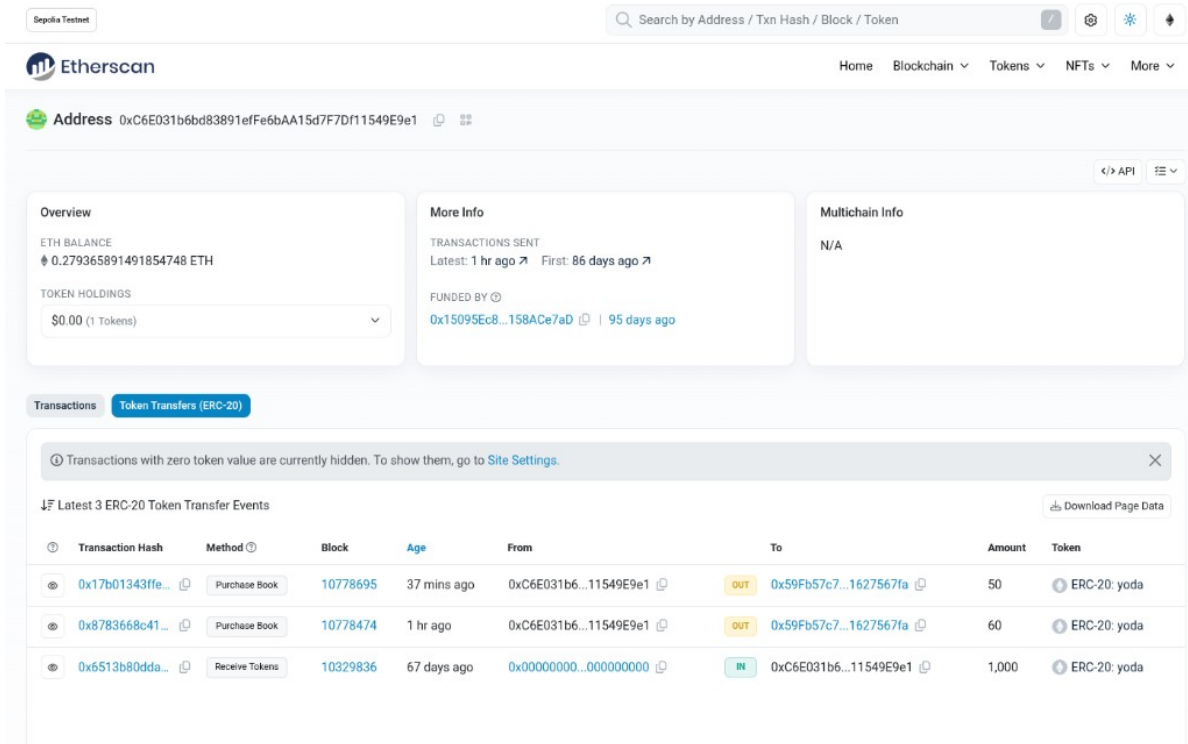


Figure 7: Yoda ERC-20 token transfers from the seller wallet.

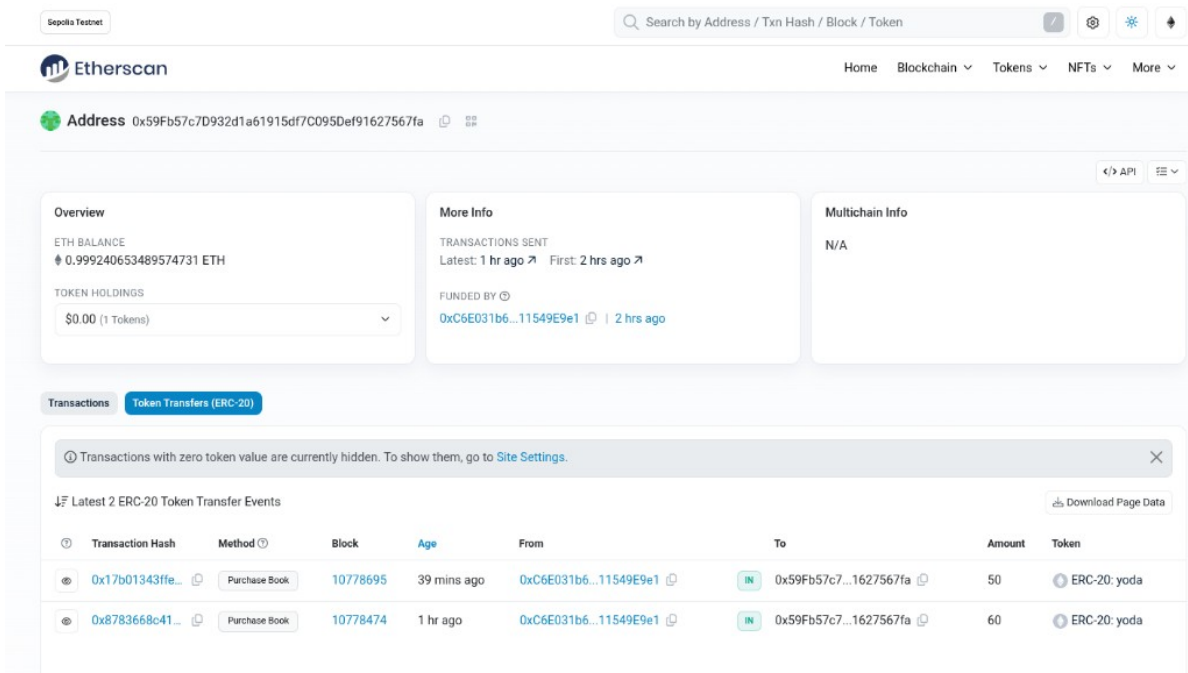


Figure 8: Yoda ERC-20 token transfers received by the buyer wallet.

Use of AI Tools

Cursor (an AI-assisted IDE) was used as a development aid for code refactoring and documentation drafting. Project architecture, contract design, deployment, and testing were directed by the student.